Research Question & Hypothesis



ACE

CYBER R&D Program

Conversational AI Interface for
Analyst-Driven Threat Investigation

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Template

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1.0 Research Question

How can a Conversational AI interface specifically integrated with Elasticsearch be designed to enable SOC analysts to query log data using natural language for common adversarial threat-hunting tasks (e.g., failed logins, suspicious IPs, lateral movements)?

2.0 Hypothesis

If a Conversational AI interface is developed and integrated with Elasticsearch to support natural-language threat-hunting queries, then, when evaluated against a pre-tagged dataset containing failed-login and lateral-movement events, it will achieve an overall F1 score of at least 0.95 across all evaluated threat-hunting tasks, indicating balanced precision and recall in detecting threat-tagged log events.

3.0 Variables / Factors to Consider

3.1 Independent Variables

- Threat Hunting Task Condition (Categorical):
    - Failed Logins
    - Suspicious IPs
    - Lateral Movement
- Conversational AI Interface Condition (Categorical):
    - Manual Query
    - Conversational AI Interface Query

3.2 Dependent Variables

- Query Latency (Quantitative, time from analyst action to receipt of final results)
- Detection accuracy (Quantitative, percent of correct identifications of malicious events)

- Early Detection (Quantitative, time between the first malicious event detected and when the analyst correctly identifies it via manual methods)

3.3 Control / Contextual Factors

- Integration with Elasticsearch
- Use of a synthetic pre-tagged log dataset
- Limited to two different threat-hunting tasks
- The proficiency of the analysts

4.0 Rationale

The hypothesis is supported indirectly by converging evidence across multiple areas of the surrounding literature.

First, applied studies demonstrate that AI can enhance precision and reduce delays in threat detection. Villegas-Ch et al. achieved 95% precision with explanations in under 500ms, cutting analyst validation time by 33% [1]. Guan et al. showed that AI-driven defense frameworks reduced response delay to under five seconds while sustaining 95% detection accuracy [2]. Similarly, Ramesh et al. demonstrated that ensembles of deep learning models with metaheuristic optimization achieved over 98% detection accuracy on cyberattack datasets [3], and Horváth et al. demonstrated the effectiveness of scalable anomaly detection methods for real-time log analysis [4]. Together, these findings demonstrate the technical feasibility of achieving a latency of less than 5 seconds and an accuracy of more than 90% with benchmarks in the present hypothesis.

Second, theoretical and review-based literature emphasizes the broader role of AI in cybersecurity. Samtani et al. argue that AI is central to the long-term roadmap for vulnerability management and threat prioritization, requiring interdisciplinary approaches [5]. Gupta et al. discuss the dual-use nature of generative AI, highlighting both its defensive applications in threat intelligence and its risks, which must be mitigated when deploying conversational AI in SOCs [6]. Likewise, Dhanushkodi and Thejas highlight how explainability and resilience are essential for trustworthy AI-enabled detection, especially for large-scale, real-time environments [7]. These perspectives justify the project's focus on balancing usability, resilience, and operational value in SOC contexts.

Third, analyst trust, transparency, and usability emerge as critical human factors. Roch [8] and Hagen et al. [9] both show that analysts often distrust automated alerts, with 65% reporting skepticism due to false positives, automation bias, and poor explainability. Both

stress that hybrid, user-centric models, where analysts remain in control, are more likely to succeed. Zheng et al. further reinforce this from a human–computer interaction (HCI) standpoint, showing that conversational interfaces improve usability, reduce cognitive workload, and enhance adoption when designed with user experience in mind [10]. A conversational AI interface for SOCs directly addresses these needs by providing analysts transparent control while benefiting from AI enhancements.

Finally, Elasticsearch-centered studies demonstrate direct relevance to SOC workflows. Alsharabi et al. showed that Elasticsearch combined with advanced query languages can detect adversarial lateral movement effectively [11], while Bhardwaj et al. extended this to detecting privilege escalation through registry modification and process injection [12]. Both highlight Elasticsearch's effectiveness in threat hunting but also highlight its reliance on specialized query syntax, which limits accessibility. By introducing natural language interfaces, the present research makes these powerful detection capabilities more accessible to analysts without requiring advanced technical expertise.

Taken together, this literature suggests that a conversational interface integrated with Elasticsearch is both technically achievable (latency and accuracy benchmarks are realistic) and theoretically justified (human factors research shows analysts prefer interactive, explainable tools). Therefore, it is reasonable to expect that the hypothesis will hold true.

5.0 References

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6.0 Change Log

Ver.	Date	Description of Change	Author
v1.1	2025-09-24	Filled out document (first time)	Koi McFarland
v1.2	2025-09-27	Updated to include extra sources	Koi McFarland
v1.3	2025-11-19	Updated hypothesis and formatting	Koi McFarland
v1.4	2025-11-20	Slightly changed hypothesis	Koi McFarland

7.0 Mentor Review / Approval

Reviewer	Role	Date	Notes
Student Peer	PI	2025-12-17	Final Deliverable
Faculty Advisor			
DoD/ACE Mentor			